

Does Controlled Ownership affect Board Independence?

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Abstract

This study examines the relationship between the controlled shareholders and board independence in the context of concentrated ownership. Analysis of 7,624 firm year observations covering over 1,132 NSE listed Indian firms for the period 2001-2011 using Feasible Generalized Least Square (FGLS) estimators provide empirical evidence that there is a non-monotonic inverse U-shaped relationship between controlled shareholding and independent directors. The observed inverted U shaped relationship is consistent with the view that the interest of controlling shareholders is aligned with the minority shareholders, as the independent directors in firm's board increases with increase in controlled shareholding to safeguard the interest of minority shareholders. However, this relationship turns at the level of 38.37% of ownership, at which the alignment of interest is suppressed by the entrenchment effect. At higher levels of shareholdings, controlling shareholders maintain their discretionary power in board, which tend to discourage the presence of independent directors in the board. With the higher levels of shareholding, controlling shareholders get full control on the decision making process and can expropriate dispersed shareholder by their self- serving behaviour. This result can be viewed as the evidence towards entrenchment hypothesis and it is consistent in both pre-crisis and post-crisis period. However, in case of duality leadership structure, controlling shareholders are willing to forego their discretionary power to avoid the potential price discount. We find evidence of inverse U-shaped relationship for private and government owned firms, while controlled shareholding is unrelated to the proportion of independent directors for family and foreign owned firms. The results are robust, since we have addressed the endogeneity issue by using instrumental variable approach. Our findings support the conjecture that controlled ownership affects the board composition and supports the theory of alignment and entrenchment.

Keywords: Ownership Structure, Board Composition, Agency Problem, Independent Director, Leadership Structure, Corporate Governance

I Introduction

Governance of firms with controlled ownership may have varying impact on the board composition and it also differs due to the blockholder having controlling interest and CEO/Chair duality leadership structure. In the presence of controlled ownership and control, the nature of agency problem also differs from that of dispersed ownership. Since, minority shareholders are vulnerable to expropriation from controlling shareholders (entrenchment hypothesis). The major conflict occurs between the controlling inside shareholders that take self-serving actions and extract private benefits for their own interest in detriment to the interest of the minority shareholders (Morck & Yeung, 2004). The owners of various business groups also control companies by holding key management position and control the board by placing significant number of insiders in the board. Thus, the presence of independent directors in a board can solve the agency problem of the controlled shareholders and minority shareholders. However, if the controlled shareholders want to continue to enjoy their discretionary power over any decision in board, they will discourage the presence of independent directors in board. While there is ample evidence of the impact of insider ownership on board composition in developed economies, there are very few studies which explore the impact of controlled ownership in the context of emerging markets like India, where business is dominated by the controlled shareholders. As highlighted by Claessens & Yurtoglu (2013), there is a need for further study in emerging markets in context of family-owned and business groups. This study aims to fill this gap and contribute to literature by examining the relationship between the controlling shareholders and the proportion of independent directors on the board. We also examine the moderating impact of leadership duality structure and presence of blockholders such as government, private, family and foreign on the association between controlled shareholders and board independence. India represents an ideal setting to examine this relationship. Since, Security Exchange Board of India (SEBI) revised the rules and regulations of corporate governance and compel firms to have at least fifty percent of independent directors on the board and thirty-three percent in case chairman is a non-executive director (SEBI, 2005). In the light of this exogenous shock and the unique nature of agency problem it is pertinent to examine if agency factor impacts the choice of board composition of firms.

The ownership structure and board structure are considered to be the key corporate governance mechanism. The variation in ownership and board structure across the country results in differences in their corporate governance practices (Li, 1994). The ownership structures of the firms determine the nature of governance problem. In case of dispersed ownership, major conflict arises between the manager (agent) and the shareholders (principal) of the firms (Agency Type I) (Berle & Means, 1932) and in context of concentrated ownership, conflict arises between the controlling shareholders and dispersed minority shareholders (Agency Type II) (Roe, 2004). The agency problem between the manager and shareholders is generally observed in developed countries like US and UK (outsider system) while the agency problem between controlling shareholders and minority shareholders is noticed in Germany, Japan and Asian economies (insider system). Due to distinct nature of the agency problem, the corporate governance mechanism to control and monitor the problem also differs.

In case of outsider system, the board is responsible for aligning the interest of managers and the shareholders and largely acts as a monitoring authority towards the managers. Further in the context of insider system, the board's task is to control the expropriation of minority shareholders by controlling shareholders. This type of problem is common in countries like continental Europe, Japan and many Asian economies including India (Sarkar & Sarkar, 2012). Indian firms are largely dominated by the family firms and concentrated ownership structure. In India 93% of the listed firms have atleast one shareholder who owns 5% or more equity and 43% of firms have atleast one shareholder who owns more than 20% of share in a firm (Sarkar & Sarkar, 2012). Therefore, the free-rider problem associated with the shareholders in monitoring the manager is insignificant in this context. Nevertheless, the problem between the controlling shareholder and minority shareholders is significant. The limited discourse, lack of enforcement, presence of pyramiding and crossholding (Khanna & Palepu, 2000) discourages the minority shareholders to monitor and represent their problems. In this scenario, the role of board of directors will be crucial in protecting the interest of the minority shareholders. Specifically, there is a worldwide consensus that the independent board will be more effective (Lorsch & MacIver, 1989; Zahra & Pearce, 1989).

Independent board performs its monitoring and control role better than its counterpart. Weisbach (1988) finds that independent board is more likely to fire the underperforming

CEO. Anti-takeover provision is another important decision where conflict of interest arises. Cotter, Shivdasani, & Zenner (1997) shows that firms with majority of independent directors in a board get higher stock return compared to other firms. Board independence also plays a critical role in dealing with the situation of financial distress and crisis. Daily (1995) recorded that the board independence is positively related to successful reorganization activity of the firm. A firm with more independent directors in the board also have lower probability of being sued (Helland & Sykuta, 2005). The independent directors also perform the role of an arbitrator in case of any internal disagreement. Apart from monitoring and control function, board also performs an important function of providing strategic guidance to the firms. The board brings the external expertise and give crucial guidance to the management in forming the strategy of the firm (Zahra & Pearce, 1989). Thus, the presence of independent directors in a board can solve the agency problem of the controlled shareholders and minority shareholders and this should be linked to the level of ownership concentration. However, if the controlled shareholders want to continue to enjoy their discretionary power over any decision in board, they will discourage the presence of independent directors in board.

The existing literature in the field of ownership structure and board composition is limited to the developing economies, focusing mostly on managerial ownership and board composition (Agency Type I perspective) (Denis & Sarin, 1999; Mak , & Li, 2001; Coles, Daniel, & Naveen, 2008; Linck, Netter, & Yang, 2008). Few studies have focuses on agency Type II perspective by focusing on concentrated ownership and board composition (Mak & Li, 2001; Donnelly & Kelly, 2005; Baglioni & Colombo, 2013; Fraile & Fradejas, 2014). However, the relationship between ownership structure and board composition calls for the empirical study, especially in the context of an emerging economy like India, where business is dominated by the controlled shareholders. The nature of agency problem also differs in the Indian context, largely between the controlling shareholders and minority shareholders. India as the fastest growing economy (IMF) and as a centre of investment of the global investors in the backdrop of the corporate governance reforms provides us a relevant context for empirical study.

To the best of your knowledge there is only one study in context of India (Chauhan et al., 2016) which analyse the relationship between controlled ownership and board independence. However, our study extends this study in following ways first, by considering the leadership duality structure in the analysis and its moderating impact with controlling shareholders. Second, we strengthen the analysis by examining the impact of presence of blockholders such

as government, private, family and foreign on the association between controlled shareholders and board independence. Third, we also analysed the impact of the type of controlled shareholders: Indian or foreign and the impact of the global financial crisis period on the relationship. Fourth, we have used an improved methodology of Feasible Generalized Least Square (FGLS) estimators which gives robust results in case of heteroskedasticity and autocorrelation in the data. We have also used instrumental variable approach to address the endogeneity issue. This paper will contribute to the international literature, which is primarily on the developed economies by examining the relationship between the controlled shareholders and board independence in the Indian context of concentrated ownership. We analyse the non-monotonic relationship of controlled shareholders and board independence from the perspective of alignment of interest and entrenchment hypothesis.

We perform an in-depth analysis using Feasible Generalized Least Square (FGLS) estimators on a sample which consists of 7,624 firm's year observations covering over 1,132 NSE listed Indian firms for the period 2001-2011 and spanning across different industries. We find the non-monotonic inverse U-shaped relationship between the controlled shareholding and independent directors. The independent directors in firms' board increases as controlled shareholding increases, upto 38.37% and then falls as controlled shareholding increases. The result backs the theory of alignment of interest and supports the notion that at lower level of controlled ownership (upto 38.37%), the interests of controlled shareholders and minority shareholders is aligned, beyond which we find the evidence towards entrenchment hypothesis i.e. at higher levels of controlled ownership, majority shareholders expropriate minority shareholders and tend to discourage the presence of independent directors in board. The results are consistent in both pre-crisis and post-crisis period. We also find evidence of inverse U-shaped relationship for private and government owned firms, while controlled shareholding is unrelated to the proportion of independent directors for family and foreign owned firms. We further find that, the firms with duality leadership structure tend to have 18.56% more independent directors in the board compared to separate leadership structure. After using firm age as an instrument, the results of 2SLS also confirm an inverse U-shaped non-monotonic relationship between the controlling shareholders and independent directors. Thus, our study contributes to the existing literature on alignment theory and entrenchment theory and shows that presence of independent directors depends upon the level of the shareholding of control shareholders.

II Theoretical Framework

Ownership Structure and Agency Problem

Insider ownership (Jensen & Meckling, 1976) and board independence (Shleifer & Vishny, 1997) are considered to be major governance mechanism to control the agency cost. The agency problem was first highlighted by Berle and Means (1932) in the context of US market where ownership is dispersed. Here agency problem arises between the shareholder (principal) and manager (agent) due to the separation of ownership and control, which is also known as Type I or vertical agency problem. The basic reason of such problem is the presence of information asymmetry between the owner and manager; managers are having information advantage about the company performance and future prospects that leads to problem of moral hazard between the two. Jensen and Meckling (1976) suggest that managerial ownership is one of the effective internal governance tool to control the Type I agency problem. The participation of management in the ownership will help owners to align management interest to their interest and minimize the agency conflict. However, alignment hypothesis is less relevant in case of Type II agency problem, which arises between the controlling inside shareholders and diffused minority shareholders (Roe, 2004).

In continental Europe, Japan and Asia, the shareholding of the firms is generally concentrated and controlled by few group firms and families. In the presence of concentrated ownership and control, the nature of agency problem also differs from that of dispersed ownership. Here minority shareholders are vulnerable to expropriation from controlling shareholders (entrenchment hypothesis). The major conflict occurs between the controlling inside shareholders that take self-serving actions and extract private benefits for their own interest in detriment to the interest of the minority shareholders (Morck & Yeung, 2004). The owners of various business groups in Asian region control companies by holding key management position and control the board by placing significant number of insider in the board. The structural devices such as dual class shares and stock pyramids are used by controlling shareholders to get control rights even in case where they do not have control through cash flow rights. Therefore, with regard to the presence of controlling shareholders, their expropriation nature towards minority shareholders and irrelevance of internal governance mechanism of managerial ownership, we need an empowered and independent board to safeguard the interest of the minority shareholders.

Board Composition and Corporate Governance

The board of directors of the company performs critical roles of strategic guidance and monitoring of the management (Forbes & Milliken, 1999; Coles, Daniel, & Naveen, 2008). The composition of the board taking into consideration its function can be grouped into two major categories of inside (executive) and outside (non-executive) directors. The stewardship theory of board of directors values the presence of inside director in the board because they are better informed about the company and its strategic initiative (Helmer, 1996; Carpenter & Westphal, 2001). Their operational and industry expertise will empower them to make a significant contribution to the firm performance. Empirical study by Kesner (1987) also finds the positive relationship between the executive director and return to investment. On the other hand, from agency perspective (Johnson, Daily & Ellstrand., 1996; Sundarmurthy & Lewis, 2003) the presence of outside directors being non-executive facilitate the monitoring of the insiders and safeguard the interest of the minority shareholder in case of conflict between the controlling shareholders and dispersed shareholders. Several empirical findings support the outsider directors presence in the board and found significant and positive relationship with firm performance (Rosenstein & Wyatt, 1990; Pearce & Zahra, 1992). However, companies rationally choose board structure which is most suitable for them after considering their specific characteristics (Boone et al., 2007; Coles, Daniel, & Naveen, 2008; Lehn, Patro, & Zhao, 2009).

Ownership Structure and Board Composition

The ownership structure of the company determines the nature of the corporate governance problem and board of directors act as an effective governance mechanism. Therefore, there is indirect relationship between the ownership structure of the firm and board composition. Hence, in this study we examine the relationship between the controlling shareholders and independent directors. If agency theory holds in this context, it would be reasonable to expect that at lower levels of controlled shareholding, the interests of controlling shareholder and minority shareholders are aligned and firms will have lower number of independent directors on board. With higher levels of controlled shareholding, majority shareholders get full control on the decision making and they can expropriate minority shareholders by their self-serving behaviour. Thus, to safeguard the interest of dispersed shareholders and to minimize the expropriation, firms should have more number of independent directors on board. So, on considering both alignment of interest and entrenchment effects, we expect the relationship

between controlling shareholders and independent directors to be non-monotonic and hence we hypothesize that

H1: There is a non-monotonic relationship between the controlling shareholders and the proportion of independent directors on the board.

The presence of blockholders in firms provide better monitoring (Berle & Means, 1932) as they hold substantial stake in the firms which is backed by voting power (Jensen & Meckling 1976). Blockholders who have large shareholdings in the firms have more incentives in performing the monitoring function (Sheilfer & Vishny, 1986). Admati, Pfleiderer, & Zechner (1994) find that presence of blockholders ensures better monitoring in the firm. The presence of blockholders can solve the free rider problem of the minority shareholders and act as a mechanism to control the discretionary behaviour of the controlling shareholders. So, based on the previously discussed argument of alignment of interest and entrenchment effects, the presence of blockholders may impact the relationship between controlling shareholders and independent directors. Hence, we examine if the non-monotonic relationship of controlling shareholders and independent directors varies or remains the same for each blockholder categories, namely, government, private, family, and foreign and propose the following hypothesis:

H2: There is a non-monotonic relationship between the controlling shareholders and the proportion of independent directors on the board for different types of blockholders.

Leadership and Board Independence

The CEO/Chair duality structure allows firms to centralize management power and board leadership in one person's hands (Jensen, 1993). When the same person is appointed as a CEO and Chairman of the board, it decreases the effectiveness of monitoring function of the board (Tuggle et al., 2010). The duality structure increases the potential conflict of interest between the controlling shareholders and minority shareholders. It empowers the controlling shareholders to gain the decision making power over board and allows taking decision even at the expense of minority shareholders. Recher and Dalton (1991) also found that the firms with duality structure underperform as compared to firms with separate leadership structure. Given that CEO/Chair duality adversely impacts the board monitoring function and increases the conflict of interest. With high controlled shareholding, we expect controlled shareholder

to have a strong influence on the board in firms where same person is a Chief Executive Officer and the Chairman of the Board. Thus, board independence and other power control mechanisms will become more crucial and important to control the entrenchment of minority shareholders. So based on this argument we hypothesize as follows:

H3: The impact of interaction effect of Leadership (CEO/COB) and controlling shareholders on the proportion of independent directors on the board is positive.

III Data and Methodology

Our initial sample consists of all National Stock Exchange (NSE) listed firms. NSE since its inception in the year 1992 remains most active stock exchange in India and currently it is India's largest stock exchange. After dropping banks and utility firms and on the basis of availability of data, the final sample comprises of 7,624 firms year observation covering 1,132 NSE listed firms for a time period of 2001 to 2011, spanning across different industries. We obtain firm level financial and ownership data from Prowess database managed by Centre for Monitoring Indian Economy (CMIE). Director level data for Indian firms is hand collected data obtained from the audited annual reports of each firm for each year.

To analyse the relationship between controlling shareholders and independent directors, we used ratio of independent directors in a board to total directors in a board as the dependent variable in our analysis (Booth, Cornett & Tehranian, 2002; Fraile & Fradejas, 2014).

The main explanatory variable used in the analysis is controlled shareholding which is defined as the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies. According to the prowess database we consider the shareholding of promoters which is defined as "any person who is directly or indirectly in control of the company". In India, generally corporations have concentrated ownership and control where insiders are the controlling shareholders, different from the widely held firms where professional managers are considered as insiders.

In addition to these variables, based on the prior literature (Booth, Cornett & Tehranian, 2002; Fraile & Fradejas, 2014) our study also considers following firm level control variables in the analysis: Firm size (FSIZE) is defined as natural logarithm of total assets, there is a positive relationship between the firm size and board structure (Coles, Daniel, & Naveen, 2008; He & Sommer, 2010) and board structure helps firms to minimize the agency cost. Leverage (LEV) is measured as total borrowings to total assets, level of debt in a firm also acts as one of the monitoring mechanisms (Fraile, & Fradejas, 2014). High level of debt controls the entrancement behaviour of the management and improves utilization of resources (Jensen, 1986). Highly levered firms will have less need of board independence. Cash Flow (CFLOW), the ratio of profit after tax before depreciation and amortization by total assets, is a measure of profitability of a firm. Profitable firms tend to have more available cash and resources for controlled shareholding to expropriate and they will ensure that there discretionary power over cash flows remains intact. So, we expect that high profitable firms will not encourage board independence. Board Size (BSIZE) is measured as the lag of log of total number of directors in a board, as pointed out by Alexander, Fennell & Halpern, (1993) and Ocasio, (1994) the variation in board size itself can cause the variation in the board structure and it is relevant to consider board size as a control measure in analysing the relationship in corporate governance research. Institutional investors (INST) is a percentage of the shareholding held by the institutional investors such as mutual funds/ UTI, banks FIs and Insurance companies, FIIs, venture capital funds, foreign venture capital fund and other institutions. The change in institutional ownership over time enhances firm-level governance (Agrawal, Erel, Ferreira and Matos, 2011). The presence of institutional investors encourages firms to have more independent board. Post (POST) is a dummy variable measured as equal to 1 from the year 2005 to 2011 indicating post- mandate period and 0 from the year 2001 to 2004 indicating the pre-mandate period. Post is included in the analysis to control for the Clause 49, which is a corporate governance mandate of India implemented by Security Exchange Board of India (SEBI), a governing authority of all listed companies in India, in the year 2005. Leadership (CEO/CHAIR) is measured as 1 if the Chief Executive Officer is the Chairman of Board; else 0. Duality leadership structure of a firm decreases the effectiveness of monitoring function of the board (Tuggle et al., 2010). So, board independence will become more important to control the agency problem. To analyse the impact of type of blockholders, we identify the blockholders by using the definition of La Porta, Lopez-De-Silanes, Shleifer and Vishny (2002), who define blockholders as shareholders who holds more than 20% of the shareholding in the firms. We identify the blockholders into four broad

categories (i) government firms which are central or state government-owned corporations (ii) the family firms which are family owned firms, (iii) private firms which are non-government and non-family owned firms (iv) foreign firms which are owned by foreign origin promoters.

The panel data methodology is used to test the hypotheses. Panel data allows us to account for the firm-specific unobserved heterogeneity in the data. We began our analysis by testing the suitability of the model. We ran Hausman specification test to check for random effects and fixed effects model, based on Hausman test result we choose fixed effects models for our analysis. Industry and year dummies are included in the analysis to account for the industrial and time-related effects.

We have also run various tests to check for the model misspecification problem. We use Modified Wald heteroskedasticity test and Wooldridge test for identifying the group-wise heteroskedasticity and autocorrelation in the panel data. Both the test confirm the presence of heteroskedasticity and autocorrelation in the data. Therefore, to overcome these problems we used the Feasible Generalized Least Square (FGLS) estimates. The FGLS estimate allows us to control for the earlier mentioned problems. We test our hypothesis by using following model:

$$\begin{aligned} \text{IND_DIR}_{i,t} = & \beta_0 + \beta_1 \text{CON_OWN}_{i,t} + \beta_2 (\text{CON_OWN}_{i,t})^2 + \beta_3 \text{CEO/CHAIR}_{i,t} + \beta_4 \text{CEO/CHAIR}_{i,t} * \text{CON_OWN}_{i,t} + \beta_5 \text{INST}_{i,t} \\ & + \beta_6 \text{INST}_{i,t} * \text{CON_OWN}_{i,t} + \beta_7 \text{POST}_{i,t} + \beta_8 \text{BSIZE}_{i,t} + \beta_9 \text{FSIZE}_{i,t} + \beta_{10} \text{LEV}_{i,t} + \beta_{11} \text{CFLOW}_{i,t} + U_i + V_t + \varepsilon_{i,t} \dots\dots\dots (1) \end{aligned}$$

where IND_DIR is the main independent variable represented as the ratio of independent directors in a board to total directors in a board and CON_OWN is the main independent variable measured as the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies. CEO/CHAIR, INST, POST, BSIZE, FSIZE, LEV and CFLOW are control variables are defined previously in detail.

To analyse the effect of controlled shareholding, and its type of relationship (monotonic or non-monotonic) with the board independence we have used the CON_OWN and its quadratic term CON_OWN^2 in the analysis. To analyse the interaction effect of leadership (CEO/COB), institutional investors (INST) with controlled shareholding (CON_OWN) we

have used two interaction terms in the analysis CEO/COB * CON_OWN and INST* CON_OWN respectively.

IV Results

Table 1 represents the descriptive statistics of the sample firms. On an average, controlling shareholders hold 48.71% of the firm's equity which shows the presence of their controlling stake in the firms. On the other hand, the average institutional investor's shareholding is 12.30 % for the sample period. On an average, board consists of 9 members of which 35% are independent directors while 31.73% of board have the duality board leadership, where Chief Executive Officer is the Chairman of Board. It shows that on an average firm maintain board with one third of its directors as independent directors and majority of firms have separate leadership structure. The correlation matrix in Table 2 shows that all variables are significantly related to the independent directors. Controlled shareholdings, institutional shareholdings, board size, firm size and leverage are positively related while cash flow is inversely related to independent directors. The correlation among independent variable shows that there is no problem of multicollinearity among the variables.

To analyse the relationship between independent directors and controlled shareholders in more detail, Table 3 shows the differences in independent director's ratio for different level of controlled shareholding. We can observe that as the shareholding of controlled shareholders increases, the percentage of independent directors also increases upto 40 % level of shareholding and then there is a noticeable shift in the trend as the level of control shareholding increases to majority and super majority the percentage of independent directors shows a declining trend. It shows the possibility of presence of non-monotonic relationship between the independent directors and controlled shareholders.

Impact of controlled shareholdings on proportion of independent directors

Table 4 exhibits the results of fixed effects and FGLS analysis. Model 1 to 7 presents the results of fixed effects and Model 8 to 10 reports the results of FGLS. The results show that board independence and controlled shareholding are interrelated. The co-efficient of CON_OWN is positive and significant in all models while the co-efficient of CON_OWN² is negative and significant, suggesting a non-monotonic inverse U-shaped relationship between

the controlled shareholding and independent directors . Differentiating the quadratic equation of independent directors and control shareholding and equating it to zero, we obtain the level of controlled shareholding for which independent directors are maximum, after which independent directors tend to decrease. The independent directors in firms' board increases as controlled shareholding increases, upto 38.37% in model 10 (40.40 % in model 4) and then falls as controlled shareholding increases. The turning point of the relationship stands at 38.37% of controlled shareholding.

The positive and significant coefficient of CON_OWN shows the positive relationship between the independent directors and controlled shareholding. The positive relationship confirms the presence of significant independent directors in the board as the controlled shareholding increases to safeguard the interest of dispersed shareholders. This result backs the theory of alignment of interest and support the notion that at lower levels of controlled ownership, the interests of controlled shareholders and minority shareholders are aligned. Moreover, the negative and significant coefficient of CON_OWN² suggests that controlling shareholders expropriate at higher levels of shareholdings and to continue to enjoy their discretionary power over any decision in board, they tend to discourage the presence of independent directors in the board. This result can be viewed as the evidence towards entrenchment hypothesis. With the higher levels of shareholding, controlling shareholders get full control on the decision-making process and can expropriate dispersed shareholder by their self- serving behaviour. We call this interesting phenomenon of an inverse U-shaped non monotonic relationship between controlled shareholding and independent directors as expropriate effect / supremacy of majority, where the number of independent directors in boards increases as controlled shareholding increases and this relationship turns at 38.37% and then presence of independent directors in boards starts falling as controlled shareholding increases.

Booth, Cornett & Tehranian (2002) in there linear regression analyses find the negative relationship between the outside directors and inside ownership. However, results of Fraile & Fradejas (2014) differ as they find a non-linear relationship between the independent director and insider ownership. Similar to Fraile & Fradejas (2014), we find a non-monotonic relationship between the independent director and controlled shareholding and conclude that the level of control shareholdings in a firm influences the presence of independent directors in the board.

Table 4 further shows the findings of the impact of controlling shareholders on the independent director's changes in the presence of duality leadership structure. The co-efficient of interaction term of control shareholding and duality structure is positive and significant. If the position of CEO and Chairman is unified along with the significant presence of the control shareholding, the risk of entrenchment of the minority shareholders increases considerably, this result in greater presence of independent directors in a board to safeguard the interest of dispersed shareholders. The firm with duality leadership structure tends to have 18.56% (0.0721/0.3884) more independent directors in the board compared to separated leadership structure. The controlling shareholders are willing to forego their discretionary power in the decision making process to avoid the potential price discount, which can arise from dispersed shareholders' concern of expropriation by control shareholders.

In model 10, the control variables, POST and FSIZE are significant and positive, implying that larger firms have greater presence of independent directors in the board and firms in the post-mandate period have more independent directors compared to pre-mandate period. The co-efficient of BSIZE is negative and significant, indicating that as the board size increases, the proportion of independent director decreases. Finally, the co-efficient of INSTI, LEV and CFLOW is insignificant, suggesting that the presence of institutional investors, level of debt and profitability of the firm is unrelated to the proportion of independent directors.

Impact of Type of Blockholders on the relationship of the controlling shareholders and independent directors

In this section, we analyse the impact of blockholders on the relationship between the controlling shareholders and independent directors in a board. We repeat the same analysis for different blockholders' categories such as government firms, private firms, family firms and foreign firms and try to find whether the relationship changes with the presence of blockholders in the firms. Table 5, Model 11 to 14 show the results of government and private firms. Model 12 & 13 confirms the presence of inverse U-shaped non monotonic relationship between the controlling shareholders and independent directors and this result is consistent with the pervious results. Model 15 to 18 report the results of family firms and foreign firms and the co-efficient of both CON_OWN and CON_OWN² is insignificant,

suggesting that the controlled shareholding is unrelated to the proportion of independent directors for family and foreign firms. This might be due to their distinct characteristics, family firms, which are predominantly run by closely related family members, irrespective of their level of shareholding, control the business and plays an important role in the decision making process. The board structure of family firms is also largely controlled and directed by family members. Similar to the family firms, foreign firms which are directly or indirectly controlled by the foreign promoters, regardless of their level of shareholding, control the board.

Impact of Crisis on the relationship between the controlling shareholders and independent directors

We further analyse the impact of global financial crisis on the relationship between the controlling shareholders and independent directors. During the financial crisis, economy as a whole experienced the fall in growth rates which further got transmitted to the firms. The crisis period brings the lot of necessity of expert advice and guidance for the firms, which encourage firms to appoint independent expert in the board. The presence of independent directors in the board also act as signal to the shareholder of better governance in the crisis period. In order to find the impact of increased demand of independent directors on the relationship between controlling shareholders and independent directors, the total sample is divided into two sub-samples as pre-crisis period and crisis-affected period. Pre-crisis period is taken from year 2001 to 2007 and the crisis affected period from year 2008 to 2011. Table 6, shows the results of the analysis and confirms the presence of inverse U-shaped non monotonic relationship between the controlling shareholders and independent directors in both pre-crisis and crisis-affected period. This shows that there was no effect of crisis on the relationship of controlling shareholders and independent directors and relationship remains consistence in both the period. The result supports the finding of Kumar and Vashisht (2011) that the Indian market is not much affected by the crisis.

Impact of type of controlled shareholders on the relationship between the controlling shareholders and independent directors

In this section, we analyse the impact of type of controlled shareholders i.e. whether the controlling shareholder is from India or foreign country. The origin of the controlled

shareholders may have the impact on the relationship as foreign origin controlling shareholders may be influenced by its parent country's corporate governance and Indian origin controlling shareholders may have the experience and home advantage compared to their counterparts. Table 7 shows that the results for the Indian origin controlling shareholders are similar to previously reported results and confirm inverse U-shaped non-monotonic relationship between the controlling shareholders and independent directors. While in case of foreign origin controlling shareholders, the co-efficient of CON_OWN is positive and insignificant, this indicates that the presence of foreign origin controlling shareholders is unrelated to the presence of the independent directors in the board. This may be due to the fact that board structure of foreign origin controlling shareholders is largely affected by its country of origin and their parental firms' corporate governance practices.

V Endogeneity Test

The presence of endogeneity is always the concern in analysing the relationship between the ownership structure and corporate governance measures. The issue of whether the ownership structure defines the board structure or board structure influences the ownership structure of the firm, is a potential endogeneity problem. Durbin-Wu- Hausman and Chi2 statistic reported in Table 8, confirms the presence of endogeneity issue. This could result in the over-estimation of the results. Instrumental variable and two stage least squares model (2SLS) approach is used to deal with the endogeneity issue. The 2SLS method is more robust and superior as compared to generalised method of moments (GMM), 3SLS, etc (Miller et al., 2007).

Firm age is used as the instrument in the model and is defined as the number of years elapsed since its incorporation. As firm gets matured, demand for external capital increases which leads the firm to become more widely held. In this scenario, the controlling shareholder would reduce over time and outside shareholding will increase. The F-statistics in Table 8, which represents the strength of the instrument, shows that the instrument is robust and has an impact on the firm ownership structure. Consistent with prior results, the estimated controlled shareholdings co-efficient in the 2SLS regression (Table 8) are highly significant. The results of 2SLS also confirm our finding that there is an inverse U-shaped non-monotonic relationship between the controlling shareholders and independent directors.

VI Conclusion

Our study contributes to the existing international literature of ownership structure and board composition by extending it in the context of Indian concentrated ownership. We address the distinct issue of agency problem that is present in emerging economies and analyse the relationship between the controlled shareholding and independent directors. Moreover, we differ from the previous studies by considering the leadership duality structure in the analysis and its moderating impact with controlling shareholders in the setting of concentrated ownership and revised corporate governance mandate. Moreover, we strengthen the analysis by examining the impact of presence of blockholders and types of controlled shareholders on the relationship. We have also addressed the issue of endogeneity in the context of controlled shareholding and board independence.

We find the inverted U-shaped relationship between the controlled shareholding and independent directors. The observed inverted U shaped relationship is consistent with the view that the interest of controlling shareholders is aligned with the minority shareholders, as the independent directors in firm's board increases with increase in controlled shareholding to safeguard the interest of minority shareholders. However, this relationship turns at the level of 38.37% of ownership, at which the alignment of interest is suppressed by the entrenchment effect. The number of independent directors in board decreases with increase in controlled shareholding. At higher levels of shareholdings, controlling shareholders maintain their discretionary power in board, which tend to discourage the presence of independent directors in the board. With the higher levels of shareholding, controlling shareholders get full control on the decision making process and can expropriate dispersed shareholder by their self-serving behaviour. This result can be viewed as the evidence towards entrenchment hypothesis. Our results are in line with the Fraile & Fradejas (2014) and Peasnell, Pope & Young (2003), studies in the context of developed countries which also find the non-linear relationship between the insider ownership and outside directors. In a context of blockholders in the firms, we find the presence of inverse U-shaped relationship between the controlling shareholders and independent directors in case of government and private firms and results were consistent in both pre-crisis and crisis-affected period. The results of instrumental variable analysis also confirm the presence of non-monotonic relationship.

Overall, our study contributes to the existing literature on alignment theory and entrenchment theory and shows that presence of independent directors depends upon the level of the shareholding of control shareholders. Our study will be useful in the investing decision of shareholders, as they can check the level of the controlling shareholding as a proxy indication of the firm's board functioning and corporate governance practices. Our study has policy

implications as it highlights the need of alternative ways by which minority shareholders can be safeguarded in case of higher level of controlling ownership.

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Table 1 Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Independent Director	9957	0.3503	0.2705	0.0000	1.0000
Institutional Shareholding	9864	0.1230	0.1243	0.0000	0.8243
Controlled Shareholding	10353	0.4871	0.2073	0.0000	1.0000
Firm Size	12744	8.1459	1.6461	0.5306	13.7833
Cash Flow	12744	0.0710	0.0883	-0.9110	1.6874
Leverage	12744	0.2887	0.2024	0.0000	0.9988
Board Size	9957	9.032138	3.155437	1	30
CEO/Chair	9732	.3173	.4654	0	1

Note: The variables in the Table are measured as: Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board, Institutional Shareholding (INST): percentage of the shareholding held by the institution investors, Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets, Board Size (BSIZE): total number of directors in a board, CEO/CHAIR: 1 if the Chief Executive Officer is the Chairman of Board; else 0.

Table 2 Correlation Matrix

	Independent Dir/ Total Dir	Institutional Investors	Total Promoter	Log Total Director	Real Firm Size	Cash Flow	Leverage
Independent Director	1.0000						
Institutional Shareholding	0.0243*	1					
Controlled Shareholding	0.0618*	-0.3256*	1				
Boar Size	0.0781*	0.2448*	0.0816*	1			
Firm Size	0.1549*	0.5268*	0.1301*	0.4612*	1		
Cash Flow	-0.0361*	0.0584*	0.0299*	-0.0279*	-0.0532*	1	
Leverage	0.0322*	-0.0924*	-0.0586*	0.0136	0.0984*	-0.2569*	1

*** Significant at 0.05**

Note: The variables in the Table are measured as: Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board, Institutional Shareholding (INST): percentage of the shareholding held by the institution investors, Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets, Board Size (BSIZE): total number of directors in a board.

Table 3 Board Structure Categorized by Controlled Shareholding

Level of Controlled Shareholding	Independent Director	Board Size	Firm Size	Cash Flow	Leverage
< 1%	0.4107	8.5417	6.9680	-0.0213	0.2082
1% to 5%	0.3600	9.2542	8.5549	0.0465	0.2939
10% to 20%	0.3384	8.6453	7.8098	0.0693	0.3024
20% to 30%	0.3441	9.3578	8.7078	0.0760	0.2864
30% to 40%	0.4010	9.5165	8.6965	0.0620	0.3288
40% to 50%	0.3950	9.3679	8.5935	0.0667	0.3292
Majority -50% to 75%	0.3938	9.0137	8.3861	0.0715	0.2890
Super majority 75% & above	0.3045	9.9551	9.1782	0.0781	0.2447

Note: The variables in the Table are measured as: Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board, Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, Board Size (BSIZE): total number of directors in a board, Firm Size (FSIZE): natural logarithm of total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets, Leverage (LEV): total borrowings to total assets.

Table 4 Impact of Controlled Shareholding on independent director using fixed effects panel regression and FGLS (Indian Listed Firms, 2001-2011)

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9	Model 10
CON_OWN	0.0674** (2.53)	0.4557*** (5.36)	0.4458*** (5.23)	0.2450* (1.82)	0.6988*** (12.21)	0.6939*** (12.12)	0.5008*** (5.33)	0.7532*** (16.69)	0.7363*** (16.33)	0.3884*** (7.10)
(CON_OWN) ²		-0.4728*** (-4.81)	-0.4751*** (-4.83)	-0.3032*** (-2.22)	-0.7640*** (-12.96)	-0.7745*** (-13.10)	-0.5992*** (-7.08)	-0.8176*** (-17.91)	-0.8140*** (-17.87)	-0.5061*** (-9.91)
CEO/Chair	0.0365*** (5.01)	0.0364*** (5.00)	0.0091 (0.56)	0.0035 (0.2)	0.0451*** (7.68)	0.0187 (1.44)	0.0098 (0.71)	0.0339*** (9.34)	0.0054 (0.63)	-0.0002 (-0.04)
CON_OWN* CEO/Chair			0.0606* (1.89)	0.0701** (2.02)		0.0570** (2.27)	0.0711 (2.70)		0.0618*** (3.78)	0.0721*** (5.12)
INST				0.1968* (1.89)			0.0473 (0.71)			0.0415 (0.93)
CON_OWN*INST				-0.4556* (-1.88)			-0.1553 (-1.02)			-0.1126 (-1.10)
POST	0.5050*** (9.12)	0.5022*** (9.08)	0.5019*** (9.08)	0.5059*** (8.50)	0.4941*** (8.52)	0.4953*** (8.54)	0.5132*** (8.25)	0.4956*** (13.01)	0.4971*** (13.04)	0.5182*** (13.19)
BSIZE	0.1119*** (3.97)	0.1040*** (3.69)	0.1030*** (3.65)	0.0886*** (2.85)	-0.0430** (-2.11)	-0.0419** (-2.06)	-0.0691*** (-3.12)	-0.0023 (-0.16)	-0.0056 (-0.39)	-0.0392*** (-2.72)
FSIZE	0.0194*** (2.61)	0.0173** (2.33)	0.0177** (2.39)	0.0128 (1.48)	0.0089*** (3.64)	0.0090*** (3.67)	0.0093*** (2.84)	0.0047** (2.37)	0.0048** (2.46)	0.0051** (2.08)
LEV	-0.0500* (-1.75)	-0.0465 (-1.63)	-0.0478* (-1.68)	-0.0412 (-1.34)	-0.0011 (-0.06)	-0.0036 (-0.21)	-0.0029 (-0.16)	-0.0080 (-0.65)	-0.0098 (-0.81)	0.0029 (0.23)
CFLOW	-0.0748* (-1.74)	-0.0748* (-1.74)	-0.0764* (-1.78)	-0.0715 (-1.59)	-0.0746** (-2.01)	-0.0759** (-2.05)	-0.0773** (-2.01)	-0.0319* (-1.76)	-0.0308 (-1.62)	-0.0155 (-0.63)
Constant	-3.3217*** (-4.10)	-3.3536*** (-4.50)	-3.3502*** (-4.46)	-2.2431*** (-2.60)	-1.1850*** (-2.96)	-1.1818*** (-2.91)	-1.12478* (-1.73)	-2.21631*** (-4.84)	-2.2069 (-4.64)	-1.1130** (-2.30)
Industry Effect					YES	YES	YES	YES	YES	YES
Year Effect	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
F test	95.01	91.09	86.26	66.52	89.64	85.00	63.48			
R square	0.1169	0.1392	0.1395	0.1277	0.2310	0.2315	0.2183			
RMSE					0.2268	0.2267	0.2271			
N	7624	7624	7624	7057	7624	7624	7057	7594	7594	7007
Sigma_u	.1540	.1487	.1488	.15033						
Sigma_e	.2026	.2022	.2022	.20275						
Rho	.3664	.3510	.3513	.3547						
Wald Chi2								5305.97	5415.52	6963.47

*** Significant at 0.01, ** 0.05 and *0.10 level

Note: Values in the Table represent coefficients for each variable and *t*-statistics (for Fixed effects) and *z*-statistics (for FGLS) is in parentheses

This Table examines the Impact of controlled shareholding on the proportion of independent director for a period of 2001-2011. Model 1 to 7 are the results of fixed effects panel regression, controlling for industry & time. Model 8 to 10 are the results of feasible generalized least square (FGLS) estimator controlling for industry & time. The dependent variable is Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board. Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, (CON_OWN²): square of the percentage of controlled shareholding, CEO/CHAIR: 1 if the Chief Executive Officer is the Chairman of Board; else 0. CON_OWN* CEO/Chair: interaction term between controlled shareholding and CEO/CHAIR. Institutional Shareholding (INST): percentage of the shareholding held by the institution investors. CON_OWN*INST: interaction term between controlled shareholding and Institutional Shareholding. POST: 1 from the year 2005 to 2011 indicating post- mandate period and 0 from the year 2001 to 2004. Board Size (BSIZE): the lag of log of total number of directors in a board, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets.

Table 5 Impact of type of Blockholders on relationship between the controlling shareholders and independent directors using fixed effects panel regression and FGLS (Indian Listed Firms, 2001-2011)

Variable	Model 11	Model 12	Model 13	Model 14	Model 15	Model 16	Model 17	Model 18
	Government		Private		Family		Foreign	
CON_OWN	0.8272 (1.49)	1.0848** (2.44)	0.4179*** (3.35)	0.1463** (2.29)	-0.1258 (-0.69)	-0.0203 (-0.14)	-0.2588 (-0.48)	0.2705 (0.48)
(CON_OWN) ²	-0.8467 (-2.08)	-0.9925*** (-2.95)	-0.4634*** (-3.99)	-0.1761*** (-2.73)	-0.0077 (-0.05)	-0.0468 (-0.35)	0.0336 (0.08)	-0.3118 (-0.72)
CEO/Chair	0.1347 (1.73)	0.1193** (2.52)	-0.0035 (-0.18)	-0.0073 (-0.73)	0.0102 (0.32)	-0.0026 (-0.11)	0.0817** (2.40)	0.0421* (1.67)
CON_OWN*CEO/Chair	-0.1250 (-1.04)	-0.1186 (-1.58)	0.0913** (2.46)	0.0802*** (3.83)	0.0965 (1.62)	0.0752* (1.82)	0.2405 (1.57)	0.1430 (1.25)
INST	0.0026 (0.00)	0.0955 (0.15)	-0.0787 (-0.93)	-0.0591 (-0.99)	-0.0762 (-0.58)	0.0653 (0.67)	0.3634 (0.90)	0.5980 (1.64)
CON_OWN *INST	0.1544 (0.09)	-0.4113 (-0.31)	0.0643 (0.32)	0.0338 (0.24)	0.4326 (1.47)	0.0562 (0.27)	-1.1607 (-1.45)	-1.6531** (-2.12)
POST	0.3194 (2.72)	0.2372** (2.58)	0.5376*** (5.67)	0.4898*** (9.01)	0.5207*** (5.93)	0.5565*** (9.12)	0.4541** (2.27)	0.5375*** (3.56)
BSIZE	0.1722 (1.18)	0.0511 (0.62)	-0.0121 (-0.40)	0.0132 (0.65)	-0.0681* (-1.65)	-0.0459** (-2.42)	-0.3631*** (-4.11)	-0.2578*** (-4.26)
FSIZE	-0.0345 (-0.87)	-0.0398 (-1.60)	0.0107** (2.39)	0.0053* (1.66)	0.0072 (1.16)	0.0058 (1.15)	0.0368** (2.18)	0.0233* (1.68)
LEV	-0.0523 (-0.34)	0.0409 (0.44)	-0.0591** (-2.40)	-0.0272* (-1.69)	-0.0172 (-0.47)	-0.0162 (-0.56)	0.0112 (0.12)	0.0017 (0.02)
CFLOW	0.1266 (0.42)	0.1229 (0.74)	-0.1879*** (-3.70)	-0.0777** (-2.45)	-0.0907 (-1.29)	-0.0717 (-1.39)	0.1666 (1.12)	0.1559 (1.32)
Constant	.0540 (0.17)	.2452 (1.14)	-.1832* (-1.73)	-.1512** (-2.32)	.0320 (0.29)	.0248 (0.26)	.1528 (0.53)	-.0754 (-0.29)
Industry Effect	YES	YES	YES	YES	YES	YES	YES	YES
Year Effect	YES	YES	YES	YES	YES	YES	YES	YES
F test	2.76		32.62		30.27		4.73	
R square	0.5759		0.2204		0.3072		0.4264	
RMSE	0.1828		0.2256		0.2221		0.1850	
N	221	219	4039	3998	2335	2328	462	462
Rho								
Wald Chi2		809.09		6776.17		4591.57		1179.98

*** Significant at 0.01, ** 0.05 and *0.10 level

Note: Values in the Table represent coefficients for each variable and t-statistics (for Fixed effects) and z-statistics (for FGLS) is in parentheses

This Table examines the Impact of type of Blockholders on relationship between the controlled shareholding and the proportion of independent director for a period of 2001-2011. Model 11&12, 13&14, 15&16, 17&18 are the results of Government , Private, Family and Foreign Blockholders respectively. Model 11, 13, 15 & 17 are the results of fixed effects panel regression controlling for industry & time and Model 12,14,16 & 18 are results of feasible generalized least square (FGLS) estimator controlling for industry & time. The dependent variable is Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board. Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, (CON_OWN²): square of the percentage of controlled shareholding, CEO/CHAIR: 1 if the Chief Executive Officer is the Chairman of Board; else 0. CON_OWN*CEO/Chair: interaction term between controlled shareholding and CEO/CHAIR. Institutional Shareholding (INSTI): percentage of the shareholding held by the institution investors. CON_OWN*INSTI: interaction term between controlled shareholding and Institutional Shareholding. POST: 1 from the year 2005 to 2011 indicating post-mandate period and 0 from the year 2001 to 2004. Board Size (BSIZE): the lag of log of total number of directors in a board, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets.

Table 6 Impact of Crisis on relationship between the controlling shareholders and independent directors using fixed effects panel regression and FGLS (Indian Listed Firms, 2001-2011)

Variable	Model 19	Model 20	Model 21	Model 22
	Pre-Crisis (2001-07)		Post-Crisis (2008-11)	
CON_OWN	0.3071** (2.12)	0.2368*** (3.26)	0.7325*** (5.70)	0.6517*** (11.88)
(CON_OWN) ²	-0.3968*** (-2.98)	-0.3355*** (-5.10)	-0.8343*** (-7.37)	-0.7815*** (-15.44)
CEO/Chair	0.0291 (1.20)	0.0424*** (3.03)	-0.0007 (-0.04)	-0.0173** (-2.40)
CON_OWN * CEO/Chair	0.0719 (1.49)	0.0818*** (3.50)	0.0659** (2.17)	0.0919*** (7.09)
INST	0.0592 (0.56)	0.0631 (1.15)	0.0219 (0.26)	0.0097 (0.27)
CON_OWN *INST	-0.2780 (-1.10)	-0.1582 (-1.19)	-0.0271 (-0.14)	-0.0327 (-0.40)
BSIZE	-0.0223 (-0.64)	-0.0067 (-0.59)	-0.1199*** (-4.14)	-0.1321*** (-13.73)
FSIZE	0.0125** (2.28)	0.0072** (2.10)	0.0074* (1.87)	0.0050*** (3.32)
LEV	-0.0169 (-0.54)	-0.0168 (-1.29)	0.0250 (1.11)	0.0167* (1.75)
CFLOW	-0.1023 (-1.43)	-0.0565 (-1.22)	-0.0651 (-1.48)	-0.0140 (-0.65)
Constant	-.1374 (-1.50)	-.1288** (-2.07)	.3259*** (6.22)	.4117*** (17.32)
Industry Effect	YES	YES	YES	YES
Year Effect	YES	YES	YES	YES
F test	27.92		16.78	
R square	0.2214		0.1624	
RMSE	0.2536		0.1992	
N	3274	3130	3783	3697
Wald Chi2		111793.80		17478.09

*** Significant at 0.01, ** 0.05 and *0.10 level

Note: Values in the Table represent coefficients for each variable and t-statistics (for Fixed effects) and z-statistics (for FGLS) is in parentheses

This Table examines the Impact of type Crisis on relationship between the controlled shareholding and the proportion of independent director for a period of 2001-2011. Model 19&20 and 21& 22 are the results of Pre-Crisis and Crisis affected period respectively. Model 19 and 21 are the results of fixed effects panel regression controlling for industry & time and Model 20 and 22 are results of feasible generalized least square (FGLS) estimator controlling for industry & time. The dependent variable is Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board. Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, (CON_OWN²): square of the percentage of controlled shareholding,

CEO/CHAIR: 1 if the Chief Executive Officer is the Chairman of Board; else 0. CON_OWN* CEO/Chair: interaction term between controlled shareholding and CEO/CHAIR. Institutional Shareholding (INSTI): percentage of the shareholding held by the institution investors. CON_OWN*INSTI: interaction term between controlled shareholding and Institutional Shareholding. POST: 1 from the year 2005 to 2011 indicating post- mandate period and 0 from the year 2001 to 2004. Board Size (BSIZE): the lag of log of total number of directors in a board, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets.

Table 7 Impact of type of controlled shareholders on relationship between the controlling shareholders and independent directors using fixed effects panel regression and FGLS (Indian Listed Firms, 2001-2011)

Variable	Model 23	Model 24	Model 25	Model 26
	Indian Controlled Shareholders		Foreign Controlled Shareholders	
CON_OWN	0.3297*** (4.72)	0.2735*** (6.41)	0.0519 (0.55)	0.0402 (0.57)
(CON_OWN) ²	-0.3673*** (-5.31)	-0.2858*** (-6.46)	-0.2524** (-2.18)	-0.2137** (-2.57)
CEO/Chair	0.0330** (1.97)	0.0208** (2.20)	0.0430*** (2.76)	0.0488*** (5.04)
CON_OWN * CEO/Chair	0.0106 (0.33)	0.0235 (1.25)	-0.0030 (-0.07)	-0.0267 (-0.96)
INST	0.1023* (1.69)	0.1045*** (2.66)	0.0900 (1.27)	0.1534*** (2.72)
CON_OWN *INST	-0.1500 (-1.01)	-0.1711* (-1.76)	-0.2510 (-1.12)	-0.5152*** (-3.23)
POST	0.5130*** (8.46)	0.5181*** (14.00)	0.4941*** (3.16)	0.5438*** (3.95)
BSIZE	-0.0643*** (-2.88)	-0.0410** (-2.78)	-0.0675* (-1.69)	-0.0368 (-1.47)
FSIZE	0.0082** (2.52)	0.0066** (2.68)	0.0024 (0.41)	0.0029 (0.59)
LEV	-0.0135 (-0.72)	-0.0242* (-1.82)	0.0244 (0.68)	0.0164 (0.63)
CFLOW	-0.0897** (-2.30)	-0.0449* (-1.70)	0.1069 (1.47)	0.1066*** (7.96)
Constant	-0.0949 (-1.40)	-0.0981** (-2.12)	.0257 (0.16)	-0.0548 (-0.38)
Industry Effect	YES	YES	YES	YES
Year Effect	YES	YES	YES	YES
F test	66.30		20.69	
R square	0.2197		0.3145	
RMSE	0.2288		0.2135	
N	6997	6927	2210	2000
Wald Chi2		13475.17		6981.52

*** Significant at 0.01, ** 0.05 and *0.10 level

Note: Values in the Table represent coefficients for each variable and t-statistics (for Fixed effects) and z-statistics (for FGLS) is in parentheses

This Table examines the Impact of type of controlled shareholders on relationship between the controlled shareholding and the proportion of independent director for a period of 2001-2011. Model 23 & 24 and 25 & 26 are the results of Indian controlled shareholders and Foreign controlled shareholders respectively. Model 23 and 25 are the results of fixed effects panel regression controlling for industry & time and Model 24 and 26 are results of feasible generalized least square (FGLS) estimator controlling for industry & time. The dependent variable is Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board. Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, (CON_OWN²): square of the percentage of controlled shareholding, CEO/CHAIR: 1 if the Chief Executive Officer is the Chairman of Board; else 0. CON_OWN*CEO/Chair: interaction term between controlled shareholding and CEO/CHAIR. Institutional Shareholding (INSTI): percentage of the shareholding held by the institution investors. CON_OWN*INSTI: interaction term between controlled shareholding and Institutional Shareholding. POST: 1 from the year 2005 to 2011 indicating post-mandate period and 0 from the year 2001 to 2004. Board Size (BSIZE): the lag of log of total number of directors in a board, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets.

Table 8 Instrumental variable and two stage least squares model (2SLS) approach to deal with the endogeneity issue

Variable	Model 27
CON_OWN	6.2413*** (2.96)
(CON_OWN) ²	-5.5201*** (-3.04)
CEO/Chair	0.0644*** (3.45)
CON_OWN * CEO/Chair	0.0109 (0.29)
INST	2.2906*** (2.69)
CON_OWN *INST	-5.9374*** (-2.78)
BSIZE	-0.1703*** (-4.47)
FSIZE	0.0492*** (4.32)
LEV	-0.1111** (-2.46)
CFLOW	-0.1508*** (-2.67)
Constant	-1.3885** (-2.34)
Wald Chi2	166.57
Endogeneity Test	
Durbin-Wu-Hausman	
Chi2	10.4752
F stat	10.4729
Strength of Instrument	
F stat	20.5759
Adjusted R Square	0.9655

***** Significant at 0.01, ** 0.05 and *0.10 level**

Note: Values in the Table represent coefficients for each variable and z-statistics is in parentheses

This Table analyse the endogeneity issue using the Firm age as an instrument. The dependent variable is Independent Director (IND_DIR): ratio of independent directors in a board to total directors in a board. Controlled Shareholding (CON_OWN): the percentage of shareholding held by the founders, their family members (spouse, parents, brothers, sisters or children) and related companies, (CON_OWN²): square of the percentage of controlled shareholding, CEO/CHAIR: 1 if the Chief Executive Officer is the Chairman of Board; else 0. CON_OWN* CEO/Chair: interaction term between controlled shareholding and CEO/CHAIR. Institutional Shareholding (INSTI): percentage of the shareholding held by the institution investors. CON_OWN*INSTI: interaction term between controlled shareholding and Institutional Shareholding. POST: 1 from the year 2005 to 2011 indicating post- mandate period and 0 from the year 2001 to 2004. Board Size (BSIZE): the lag of log of total number of directors in a board, Firm Size (FSIZE): natural logarithm of total assets, Leverage (LEV): total borrowings to total assets, Cash Flow (CFLOW): ratio of profit after tax before depreciation and amortization by total assets.